Referee report on: "Fiscal Consolidations in Commodity-Exporting Countries: A Small Open Economy DSGE Perspective"

Summary:

The paper examines the macroeconomic implications of fiscal consolidation in commodity-exporting economies using a dynamic stochastic general equilibrium (DSGE) model adapted to reflect the fiscal dynamics of resource-rich economies. The model features key fiscal tools such as distortionary taxation, public spending, and external borrowing, and includes a non-produced commodity sector with exogenously determined revenue flows. The model is calibrated and estimated based on Ecuador's 2020–2025 IMF-backed adjustment framework. The authors conduct some simulations in which fiscal policies adhere to the consolidation plan, while commodity revenues are subject to plausible historical shocks.

The analysis suggests that variations in commodity revenues exert asymmetric impacts on fiscal and macroeconomic indicators. Unexpected increases in oil revenues facilitate faster deficit reduction, reduce the country's risk premium, and improve access to financing, which may reduce the commitment to fiscal discipline. In contrast, lower-than-expected revenues elevate borrowing costs, deteriorate fiscal balances, suppress private investment, and necessitate greater fiscal adjustments to meet program goals.

Comments

The paper is generally well written, and the authors demonstrate mastery of the tool used in their analysis. However, the current version requires some clarification to help readers better understand and contextualize the findings. Below are some comments that may be useful for improving the paper.

- The paper presents the two-agent model as if it were being used for the first time. It would be preferable to acknowledge and cite the earlier literature that introduced and developed this framework.
- The static optimization problem of the hand-to-mouth households is not specified, and the corresponding optimality conditions for their consumption and labor choices are absent from the exposition. These elements should be explicitly stated to ensure that the household sector is fully and transparently characterized.
- It would be useful for the authors to elaborate on the determinants of the interest rate paid by the government. In particular, the paper should clarify whether—and through which mechanisms—this interest rate is linked to the rate faced by Ricardian households and explain the economic rationale for any such connection.
- The paper would benefit from a clearer exposition of the dynamics of foreign GDP within the model. In particular, it is important to clarify whether the evolution of foreign GDP is related in any way to the dynamics of the non-stationary productivity shock and to explain the rationale for any such linkage or independence.
- More generally, the paper frequently presents optimization problems without deriving or discussing the corresponding first-order conditions. These conditions are essential for providing the reader with a comprehensive understanding of the forces driving agents' behavior and should

therefore be included and explained in the main text. Or the authors should direct the reader to the Appendix for details on the FOCs.

- All equations should be consistently numbered for ease of reference.
- The reader would benefit from a more detailed discussion of the transmission channels through which revenue from the non-produced commodity affects the domestic economy. In its current form, the model specification does not sufficiently articulate how fluctuations in commodity revenue influence the behavior of households and firms. Given the complexity of these channels, including their interactions with fiscal policy, household income, and external financing conditions, it is important for the authors to clearly outline these mechanisms and explain how the commodity revenue shock propagates through the model.
- The analysis in the fiscal consolidation section could be significantly strengthened, as the current exposition does not sufficiently clarify how fiscal instruments interact with household behavior, country risk dynamics, and commodity-price shocks. While the section reports the simulated outcomes of the consolidation program, it does not provide the reader with an adequate explanation of the underlying economic mechanisms that generate these results. As a consequence, the discussion lacks a coherent conceptual framework to guide the reader through the transmission channels at work.
- First, the role of the country risk premium in investment dynamics requires further elaboration. The simulations show a substantial response of private investment to changes in the premium, yet the mechanism is not clearly articulated. In particular, it is not evident whether this sensitivity reflects specific modelling assumptions—such as a debt-elastic external interest rate—and how this channel quantitatively influences capital accumulation.
- Second, the section should more explicitly acknowledge that fiscal consolidation simultaneously affects aggregate demand and supply in the model. On the demand side, reductions in government consumption, investment, and transfers lower domestic absorption. On the supply side, the resulting decline in the country risk premium reduces the cost of capital and stimulates private investment. Although both channels are mentioned, the text does not clearly identify them as opposing forces or discuss which mechanism dominates at different horizons. A more explicit treatment of these interactions would greatly improve the reader's understanding of the short- and medium-run macroeconomic effects of consolidation.
- Third, the implications for labor supply are insufficiently explained. In the earlier commodity-boom simulations, positive income effects from commodity windfalls reduce labor supply. By contrast, in the consolidation scenarios, the reasons behind shifts in hours worked—both for optimizing households and for hand-to-mouth households—are not well articulated. A brief explanation of the labor-supply drivers for each household type, and of why their patterns differ across scenarios, would enhance the clarity of the results.
- Fourth, the interaction of fiscal instruments needs better explanations. The consolidation package involves several components, including a permanent increase in labor income taxation, a temporary rise in capital income taxation, changes in transfers, reductions in public investment, and modest cuts in government consumption. The reader would benefit from a discussion of the relative contribution of each instrument: which elements drive the short-run contraction, which support the medium-run recovery, and how different household types are affected. A concise verbal decomposition of the macroeconomic effects associated with each fiscal tool would be highly informative.

- Fifth, the explanation of the U-shaped response of aggregate consumption could be expanded. Although the section describes the pattern—an initial decline followed by a rebound—it does not discuss the underlying determinants. The initial contraction likely reflects reduced transfers and higher labor taxation, which lower disposable income, particularly for hand-to-mouth households. The subsequent rebound presumably arises from a lower country risk premium, a recovery in private investment, and positive income effects on liquidity-constrained households. Making these channels explicit would help the reader understand the dynamics.
- Finally, the discussion of the trade balance and external sector could be improved. The model shows an improvement in the trade balance during consolidation, but the mechanism is not linked to the underlying drivers. It is unclear whether this improvement results primarily from the decline in domestic absorption, from the investment rebound under lower risk premia, or from changes in external borrowing costs. A more intuitive explanation of the transmission channels through which fiscal consolidation affects external balances would enhance the interpretation of the results.